

Granular RVEs

Three-dimensional computational mechanics of realistic granular microstructures using
FEniCSx

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September 10, 2026

Repository	jay-a/granular-rves
Source commit	3c4ee5e
Generated	2026-09-10T19:38:11.205593+00:00
Author	Jay M. Appleton
Status	Early development
Tests	8 passed / 8 total
Build	passed

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1 Introduction

This document describes the development of the *Granular RVEs* research software project, a three-dimensional computational mechanics framework for studying realistic granular microstructures using FEniCSx.

The project is being developed as a research-oriented software framework for finite element modelling of granular assemblies, with an emphasis on reproducibility, verification, validation, and extensibility toward homogenization, uncertainty quantification, and data-driven surrogate modelling.

The initial computational program progresses from single-grain mechanics to interacting grains, periodic assemblies, and realistic three-dimensional granular microstructures. Later stages will address homogenization, parameter identification, uncertainty quantification, and data-driven surrogate models.

2 Theory

This section is under development.

3 Computational Toolchain

This section is under development.

4 Usage

This section is under development.

5 Computational Experiments

This section is under development.

6 Verification and Validation

This section is under development.

7 Results

This section is under development.

8 Research Direction

This section is under development.

9 Reproducibility

This section is under development.

References